Sidharth Dutta

Atlanta, GA | sidharth.dutta1234@gmail.com

Objective

Computer Engineering student at Georgia Tech seeking software, embedded systems, chip design, or PCB design internships starting May 2027. Experience in Swift/SwiftUI application development, computational protein engineering, and technical problem solving.

Education

Georgia Institute of Technology | Atlanta, GA *August 2026 - Present*
Bachelor of Science in Computer Engineering, GPA 4.0 Expected Graduation, May 2029

Skills

Programming: Swift, Python
Frameworks / Platforms: SwiftUI, iOS, HealthKit, MapKit, CoreLocation
Software / Tools: Xcode, Git, GitHub, JSON, REST APIs
Engineering: Structure-guided protein engineering, codon optimization, plasmid design, Arduino, basic circuits
Technical Interests: Embedded systems, chip design, PCB design
Strengths: Adaptability, critical thinking, collaboration, communication, leadership
Interests: Cycling, calligraphy, fountain pens, calisthenics, photography, computer hardware

Experience

UNC Wilmington / NC School of Science and Math | Wilmington, NC *May - July 2025*
Neuroscience Computational Protein Design Researcher

- Applied structure-guided protein-engineering principles to design a Huntington's disease therapeutic platform using 2GSG-derived polyglutamine-binding regions and a humanized 3EBA scaffold.
- Designed a modular construct combining an anti-transferrin receptor delivery arm, mutant-huntingtin-binding intrabody, peptide linker, and LAMP1 lysosomal-targeting component.
- Contributed to host-specific codon optimization and plasmid design, placing anti-TfR sequences in a pET28 backbone for E. coli expression and 2GSG-linker-LAMP1 sequences in a pIVTRup backbone for human-cell expression.
- Supported plasmid production and purification; presented findings at a university symposium and co-authored a scientific research paper.

Projects

Cadence | Personal iOS Application *2026 - Present*
Developer

- Developed an offline-first workout and run tracker in SwiftUI with local JSON persistence, personal-record analytics, progress charts, and optional REST API synchronization.
- Integrated HealthKit and asynchronous actor-isolated networking; validated storage, analytics, sync, and health-data logic with 47 passing unit tests.

Beeline | Campus Navigation iOS Application *2026 - Present*
Developer

- Built a SwiftUI campus-navigation app using MapKit and CoreLocation for building, room, course, and place search.
- Implemented graph-based walking and indoor routing, floor-plan lookup, live Stinger bus data, and study-room availability; validated routing and data models with 83 passing unit tests.

Relevant Coursework

Physics: Mechanics, electromagnetism, and circuit fundamentals through calculus-based introductory physics
Linear Algebra: Matrix operations, vector spaces, eigenvalues/eigenvectors, and systems of linear equations
Chemistry: Atomic structure, bonding, stoichiometry, and chemical equilibrium
ECE Digital Design and Circuits: Boolean algebra, combinational and sequential logic design, and circuit analysis using Ohm's and Kirchhoff's laws

Leadership or Activities

American Robotics Academy | Cary, NC *June 2024 - August 2026*
Youth Engineering and Taekwondo Summer Camp Instructor

- Designed and led week-long lessons on robotics, basic circuits, and programming with LEGO and Arduino for middle school students.
- Mentored campers, adapted instruction for different learning styles, and hosted a battle bot challenge requiring students to apply engineering concepts.

Green Level Club | Cary, NC *2024 - 2026*
President, Adopt-a-Grandparent Club

- Exchanged 400+ letters with older adults through a student-led pen-pal program, fostering intergenerational connections.
- Managed 30+ recurring members and organized five events, including senior prom, Halloween, and retirement-center gardening activities.